

PROJECT DEVELOPMENT & MANAGEMENT PLAN

Wildfire Risk Prediction Using Climate & Environmental Data

Master of Data Science · Final Year Project

Student Name	Maragani Gireesh
Programme	Master of Data Science
Project Title	Wildfire Risk Prediction Using Climate & Environmental Data
Dataset	1.88M US Wildfires — Kaggle (rtatman/188-million-us-wildfires)
Start Date	26 May 2026
End Date / Deadline	15 July 2026
Total Duration	7 Weeks
Supervisor	Andy Tai Hock Lin

Project Links

GitHub	https://github.com/Gireesh462/Data-Science-Project
Google Drive	https://drive.google.com/drive/folders/1qNq5ZLyEogtSL4f5DMWdhrM3tz9bf8xu

Project Overview

Wildfires pose a serious threat to lives, ecosystems, and infrastructure across the United States. This project applies machine learning to predict whether a wildfire event is High Risk or Low Risk, using 1.88 million historical wildfire records spanning 1992 to 2015.

Four classification models are built and evaluated — Ridge Classifier, Random Forest, XGBoost, and MLP Neural Network. Class imbalance is handled using Balanced Bagging Classifier, and XGBoost is fine-tuned with Optuna hyperparameter optimisation. ELI5 explainability is applied to identify the environmental features that most strongly influence wildfire risk.

The project is evaluated using Accuracy, F1-Score, ROC-AUC, and 5-Fold Stratified Cross-Validation to ensure robust and reliable results. All work is implemented in Python on Google Colab, with code maintained on GitHub and outputs stored on Google Drive.

1. Research Questions

Main Research Question:

“To what extent can machine learning models accurately predict wildfire risk classification using historical climate and environmental data?”

Sub-Questions:

- RQ1: Which model — Ridge Classifier, Random Forest, XGBoost, or MLP Neural Network — achieves the highest ROC-AUC score for wildfire risk prediction?
- RQ2: Which environmental features are the strongest predictors of High Risk wildfire events, identified through ELI5 explainability?
- RQ3: How does Balanced Bagging Classifier improve F1-score on the class-imbalanced wildfire dataset?
- RQ4: How much does Optuna hyperparameter tuning improve XGBoost performance over its default configuration?

2. Project Objectives

- Download the 1.88M US Wildfire dataset directly from Kaggle using the kagglehub API.
- Clean and preprocess data — missing values, encoding, duplicates.
- Engineer features: SEASON, RISK_BINARY, FIRE_SIZE_LOG, REGION.
- Apply Balanced Bagging Classifier to handle the 90/10 class imbalance.
- Produce 5 EDA charts and 1 interactive geospatial Folium heatmap.
- Train 4 models: Ridge Classifier (baseline), Random Forest, XGBoost, MLP Neural Network.
- Fine-tune XGBoost using Optuna to maximise ROC-AUC.
- Evaluate with Accuracy, F1, ROC-AUC, PR Curve, and 5-Fold Stratified Cross-Validation.
- Apply ELI5 to identify top environmental predictors of wildfire risk.
- Push all code to GitHub and save all outputs to Google Drive.
- Submit full thesis by 15 July 2026.

3. Detailed Week-by-Week Plan

Week	Dates	Phase	Tasks	Deliverable
1	26 May – 1 Jun	Project Setup	Configure Kaggle API, install libraries. Download dataset via kagglehub. Auto-detect SQLite table. Inspect all columns. Create GitHub repo and Google Drive project folder.	Environment ready + dataset downloaded
2	2 Jun – 8 Jun	Data Cleaning	Handle missing values. Drop rows missing FIRE_SIZE_CLASS, LATITUDE, LONGITUDE, STATE, STAT_CAUSE_DESCR. Fill OWNER_DESCR nulls. Remove duplicates. Validate clean dataset. Push notebook to GitHub.	Clean dataset ready
3	9 Jun – 15 Jun	Feature Eng. + EDA	Create SEASON, RISK_BINARY, FIRE_SIZE_LOG, REGION. Apply Balanced Bagging. Scale with StandardScaler. Produce 5 EDA plots +	Feature matrix + 6 plots saved

Week	Dates	Phase	Tasks	Deliverable
			Folium heatmap. Save all plots to Google Drive.	
4	16 Jun – 22 Jun	Model Training	Train Model 1: Ridge Classifier (baseline). Train Model 2: Random Forest. Train Model 3: XGBoost. Train Model 4: MLP Neural Network. Record Accuracy, F1, ROC-AUC for all 4 models.	4 trained models + scores
5	23 Jun – 29 Jun	Optuna Tuning + Evaluation	Fine-tune XGBoost with Optuna. Run 5-fold stratified CV on all models. Plot confusion matrix, ROC curve, PR curve. Apply ELI5 explainability on best model. Print all classification reports. Save outputs to Google Drive.	Tuned XGBoost + eval plots
6	30 Jun – 6 Jul	Report Ch. 1–3	Save all models with joblib. Write Chapter 1 (Introduction). Write Chapter 2 (Literature Review). Write Chapter 3 (Dataset & EDA). Save thesis draft to Google Drive.	Ch. 1–3 drafted
7	7 Jul – 15 Jul	Report Completion	Write Chapters 4–7 (Methodology, Results, Discussion, Conclusion). Proofread. Final push to GitHub and Google Drive. Submit to supervisor by 15 July 2026.	FINAL SUBMISSION ✓

4. Techniques We Are Using & Why

No.	Technique	Why We Are Using This
1	Ridge Classifier	Baseline model with L2 regularisation. Simple and fast — all other models must beat it.
2	Random Forest	Handles noisy wildfire data well. Reduces overfitting. Strong on tabular structured data.
3	XGBoost	Best performer on structured tabular data. Fine-tuned with Optuna for maximum accuracy.
4	MLP Neural Network	Captures complex non-linear patterns between climate features. Represents deep learning comparison.
5	Balanced Bagging	Fixes 90/10 class imbalance by balancing each bootstrap sample automatically.
6	Optuna	Intelligently tunes XGBoost parameters — faster and smarter than Grid Search.

No.	Technique	Why We Are Using This
7	ELI5	Explains model predictions in plain English — shows which features drive High Risk predictions.
8	5-Fold Stratified CV	Tests the model 5 times on different data splits to confirm results are stable.
9	Folium + Matplotlib	Folium for interactive geographic heatmap. Matplotlib for all EDA and evaluation charts.

5. References

All references are formatted in Harvard referencing style as required by university guidelines.

Short, K.C. (2017) Spatial wildfire occurrence data for the United States, 1992–2015. Fort Collins, CO: Forest Service Research Data Archive. Available at: <https://doi.org/10.2737/RDS-2013-0009.4> (Accessed: 26 May 2026).

Goodfellow, I., Bengio, Y. and Courville, A. (2016) Deep Learning. Cambridge, MA: MIT Press. Available at: <https://www.deeplearningbook.org> (Accessed: 26 May 2026).

Chen, T. and Guestrin, C. (2016) ‘XGBoost: A scalable tree boosting system’, Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. Available at: <https://doi.org/10.1145/2939672.2939785> (Accessed: 26 May 2026).

Breiman, L. (2001) ‘Random forests’, Machine Learning, 45(1), pp. 5–32. Available at: <https://doi.org/10.1023/A:1010933404324> (Accessed: 26 May 2026).

Akiba, T., Sano, S., Yanase, T., Ohta, T. and Koyama, M. (2019) ‘Optuna: A next-generation hyperparameter optimization framework’, Proceedings of the 25th ACM SIGKDD International Conference, pp. 2623–2631. Available at: <https://doi.org/10.1145/3292500.3330701> (Accessed: 26 May 2026).

Lemaitre, G., Nogueira, F. and Aridas, C.K. (2017) ‘Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning’, Journal of Machine Learning Research, 18(17), pp. 1–5. Available at: <https://jmlr.org/papers/v18/16-365> (Accessed: 26 May 2026).

Korobov, M. and Lopuhin, K. (2020) ELI5: A library for debugging and explaining machine learning classifiers and regressors. Available at: <https://github.com/eli5-org/eli5> (Accessed: 26 May 2026).

Jain, P., Coogan, S.C., Subramanian, S.G., Crowley, M., Taylor, S. and Flannigan, M.D. (2020) ‘A review of machine learning applications in wildfire science and management’, Environmental Reviews, 28(4), pp. 478–505. Available at: <https://doi.org/10.1139/er-2020-0019> (Accessed: 26 May 2026).

Abatzoglou, J.T. and Williams, A.P. (2016) ‘Impact of anthropogenic climate change on wildfire across western US forests’, Proceedings of the National Academy of Sciences, 113(42), pp. 11770–11775. Available at: <https://doi.org/10.1073/pnas.1607171113> (Accessed: 26 May 2026).

Sayad, Y.O., Mousannif, H. and Al Moatassime, H. (2019) ‘Predictive modeling of wildfires: A new dataset and machine learning approach’, Fire Safety Journal, 104, pp. 130–146. Available at: <https://doi.org/10.1016/j.firesaf.2019.01.006> (Accessed: 26 May 2026).